



西安交通大学
XI'AN JIAOTONG UNIVERSITY

Paper Sharing — OPD&OPSD

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OPSD 的应用

□ Knowledge Distillation

<https://arxiv.org/abs/1503.02531>

Distilling the Knowledge in a Neural Network

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大模型太贵，小模型能力弱

那就让小模型不只学 hard label，而是学大模型输出的 soft distribution。

问题：这类蒸馏通常是 off-policy 的，学生学的是 teacher 轨迹，不是自己会走到的轨迹。

□ Imitation Learning / DAgger

<https://arxiv.org/abs/1011.0686>

提出问题：如果只训练 learner 模仿 expert 的完美轨迹，部署时 learner 一旦走偏，就会进入训练集中没见过的状态，然后错误累积。

A Reduction of Imitation Learning and Structured Prediction
to No-Regret Online Learning

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不要只在专家轨迹上训练学生；
让学生自己跑，学生走到哪里，就让专家在那里标注正确动作；
不断把这些新状态加入训练集。

□ RLHF / PPO:

<https://arxiv.org/abs/2203.02155>

Training language models to follow instructions with human feedback

Long Ouyang* Jeff Wu* Xu Jiang* Diogo Almeida* Carroll L. Wainwright*
Pamela Mishkin* Chong Zhang Sandhini Agarwal Katarina Slama Alex Ray
John Schulman Jacob Hilton Fraser Kelton Luke Miller Maddie Simens
Amanda Askell† Peter Welinder Paul Christiano*†
Jan Leike* Ryan Lowe*

OpenAI

到 LLM post-training, RLHF 变成主流: 模型自己 rollout, 然后 reward model 或 human preference 给分, 再用 PPO 更新。

问题: reward 通常太稀疏: 整段回答一个分数, 但不知道每个 token 到底贡献了什么。

□ DPO:

<https://arxiv.org/abs/2305.18290>

Direct Preference Optimization: Your Language Model is Secretly a Reward Model

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它解决的是 PPO 太复杂、太不稳定的问题。
DPO 直接用 chosen/rejected pair 训练，不
显式训练 reward model，也不跑复杂 RL。

问题：它简单稳定；DPO 通常还是离线数据，且信号是 response-level pairwise preference，不是 student 当前 rollout 上的 token-level supervision。

□ Reasoning RL / GRPO:

<https://arxiv.org/abs/2402.03300>

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

Zhihong Shao^{1,2*}, Peiyi Wang^{1,3*}, Qihao Zhu^{1,3*}, Runxin Xu¹, Junxiao Song¹,
Xiao Bi¹, Haowei Zhang¹, Mingchuan Zhang¹, Y.K. Li¹, Y. Wu¹, Daya Guo^{1*}

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<https://github.com/deepseek-ai/DeepSeek-Math>

不用单独 value model，而是对同一问题采样一组回答，用组内相对分数估计 advantage。这在数学推理里很自然，因为可以用最终答案/verifier 给 reward。

问题：依赖最终结果 reward。对长推理来说，你知道最后对错，但不一定知道中间哪一步该怎么改。

□ OPD:

<https://thinkingmachines.ai/blog/on-policy-distillation/>



Off-Policy Distillation:

信号密，但不在 student 自己的状态上

RL / GRPO:

on-policy, 但 reward 稀疏



For example, we may wish to train a compact model to solve math questions such as:

prompt What is $5 + (2 \times 3)$?

We can do on-policy training via reinforcement learning, by grading each student rollout on whether it solves the question. This grading can be done by a human, or by a “teacher” model that reliably gets the correct answer.

student trajectory

5

+

2

is

7

,

and

7

×

3

is

21

.

reward: 0





□ Method

python

```
# Initialize teacher client (main):
teacher_client = service_client.create_sampling_client(
    base_model=teacher_config.base_model,
    model_path=teacher_config.load_checkpoint_path,
)

# Sample trajectories (main):
trajectories = do_group_rollout(student_client, env_group_builder)
sampled_logprobs = trajectories.loss_fn_inputs["logprobs"]

# Compute reward (compute_teacher_reverse_kl):
teacher_logprobs = teacher_client.compute_logprobs(trajectories)
reverse_kl = sampled_logprobs - teacher_logprobs
trajectories["advantages"] = -reverse_kl

# Train with RL (train_step):
training_client.forward_backward(trajectories, loss_fn="importance_sampling")
```

□ Experiment

Method	AIME'24	Teacher FLOPs	Student FLOPs	CE vs SFT-2M
Initialization: SFT-400K	60%	8.5×10^{20}	3.8×10^{20}	–
■ SFT-2M (extrapolated)	~70% (extrapolated)	3.4×10^{21}	1.5×10^{21}	1×
■ Reinforcement learning	68%	–	–	≈1×
■ On-policy distillation	70%	8.4×10^{19}	8.2×10^{19}	9-30×

Method	AIME'24	GPQA-Diamond	GPU Hours
■ Off-policy distillation	55.0%	55.6%	Unreported
■ + Reinforcement learning	67.6%	61.3%	17,920
■ + On-policy distillation	74.4%	63.3%	1,800

GPQA 全称是：

Graduate-Level Google-Proof Q&A
意思是研究生级别、很难靠简单搜索解决的科学问答题。题目覆盖物理、化学、生物等领域，难度比较高。

□ OPSD:

[\[2601.18734\] Self-Distilled Reasoner: On-Policy Self-Distillation for Large Language Models](#)

Self-Distilled Reasoner: On-Policy Self-Distillation for Large Language Models

Siyan Zhao^{† 1} Zihui Xie² Mengchen Liu³ Jing Huang³ Guan Pang³ Feiyu Chen^{*, ‡ 3} Aditya Grover^{* 1}

	SFT/Off-Policy Distillation	GRPO	On-Policy Distillation	On-Policy Self-Distillation (Ours)
On-Policy Data	✗	✓	✓	✓
Dense Learning Signal	✓	✗	✓	✓
Low Sampling Cost	✓	✗	✓	✓
No External Teacher	✓	✓	✗	✓

OPD :

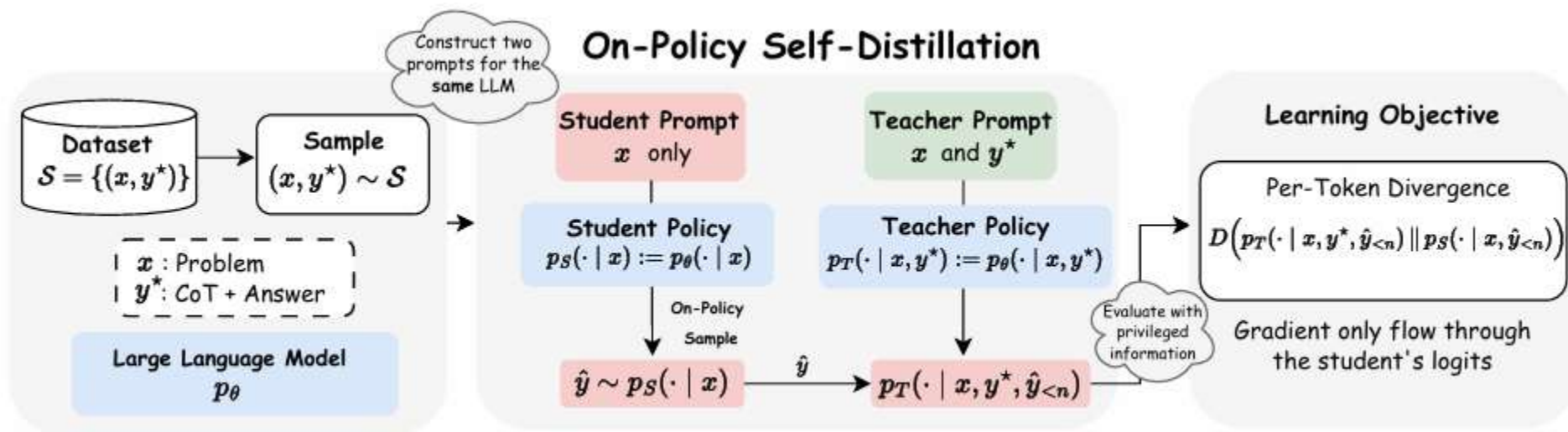
依赖外部的强大的 teacher 模型



□ Motivation:

受到人类的学习方式的启发:

当遇到问题困难时，学生不仅仅通过反复试错，而是可以查看相关教辅书或者答案，理解其中的推理，并将方法内化。



Student Prompt

Problem: Find the derivative of $f(x) = 3x^2 + 2x - 5$ at $x = 2$

Answer:

Teacher Prompt

Problem: Find the derivative of $f(x) = 3x^2 + 2x - 5$ at $x = 2$

Here is a reference solution:

First find $f'(x) = 6x + 2$, then evaluate at $x = 2$: $f'(2) = 6(2) + 2 = 14$

After understanding the reference solution, please try to solve this problem using your own approach below:

Answer:



On-Policy Self-Distillation for Large Language Models

Algorithm 1 On-Policy Self-Distillation (OPSD)

Require: Reasoning dataset $\mathcal{S} = \{(x_i, y_i^*)\}_{i=1}^N$; language model p_θ ; divergence D (e.g., JSD_β)

- 1: Let $p_S(\cdot | x)$ and $p_T(\cdot | x, y^*)$ be the same model p_θ under different conditioning.
- 2: **while** not converged **do**
- 3: Sample a minibatch $\mathcal{B} \subset \mathcal{S}$
- 4: **for all** $(x, y^*) \in \mathcal{B}$ **do**
- 5: Sample on-policy response $\hat{y} \sim p_S(\cdot | x)$
- 6: Compute the token-wise divergence along the student rollout:

$$\ell(x, y^*) \leftarrow D(p_T \| p_S)(\hat{y} | x) = \frac{1}{|\hat{y}|} \sum_{n=1}^{|\hat{y}|} D(p_T(\cdot | \hat{y}_{<n}, x, y^*) \| p_S(\cdot | \hat{y}_{<n}, x))$$

- 7: Calculate loss $\mathcal{L}_{\text{OPSD}}(\theta) \leftarrow \frac{1}{|\mathcal{B}|} \sum_{(x, y^*) \in \mathcal{B}} \ell(x, y^*)$ and update θ

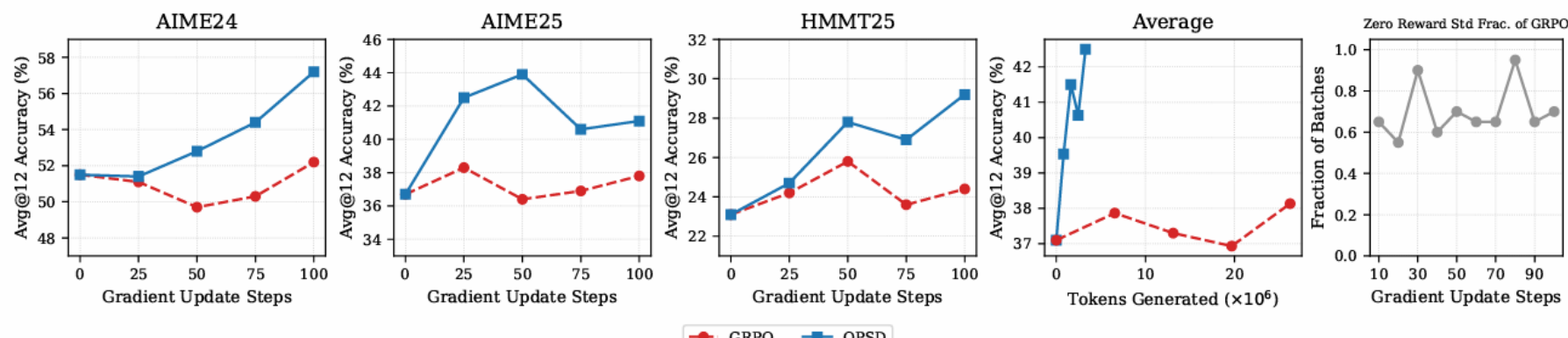
$$D(p_T \| p_S)(\hat{y} | x) \triangleq \frac{1}{|\hat{y}|} \sum_{n=1}^{|\hat{y}|} D(p_T(\cdot | x, y^*, \hat{y}_{<n}) \| p_S(\cdot | x, \hat{y}_{<n}))$$

损失函数: $\mathcal{L}(\theta) = \mathbb{E}_{(x, y^*) \sim \mathcal{S}} \left[\mathbb{E}_{\hat{y} \sim p_S(\cdot | x)} [D(p_T \| p_S)(\hat{y} | x)] \right]$

奖励函数: $A_n(x, \hat{y}) = \log p_T(\hat{y}_n | x, y^*, \hat{y}_{<n}) - \log p_S(\hat{y}_n | x, \hat{y}_{<n})$

Result:

Method	AIME24	AIME25	HMMT25	Average
<i>Qwen3-8B</i>				
Base (Instruct)	75.8	65.6	43.9	61.8
+ SFT	72.3	64.2	42.9	59.8
+ GRPO	76.4	68.9	46.7	64.0
+ OPSD	77.8	70.8	45.8	64.8
<i>Qwen3-4B</i>				
Base (Instruct)	74.9	66.4	42.2	61.2
+ SFT	70.2	62.3	43.4	58.6
+ GRPO	75.6	68.1	44.4	62.7
+ OPSD	76.4	68.3	46.1	63.6
<i>Qwen3-1.7B</i>				
Base (Instruct)	51.5	36.7	23.1	37.1
+ SFT	48.4	36.3	22.7	35.8
+ GRPO	51.1	38.3	23.7	37.7
+ OPSD	57.2	43.9	29.2	43.4



□ Ablation:

Table 3. Comparison of divergence objectives on AIME25 with Qwen3-1.7B. We report Avg@12 at different training steps. Forward KL significantly improves performance over the base model, while reverse KL and JSD ($\beta = 0.5$) show limited or negative gains.

Method	Base	Step 50	Step 100
Forward KL ($KL(p_T \parallel p_S)$)	36.7	43.9	41.1
Reverse KL ($KL(p_S \parallel p_T)$)	36.7	37.5	35.0
JSD ($\beta = 0.5$)	36.7	36.9	39.0

Method Variant	AIME25	HMMT25
OPSD w/ Full-vocabulary logit distillation (Agarwal et al., 2024)	84.1	60.0
OPSD w/ Sampled-token distillation (Lu & Lab, 2025)	82.1	57.3

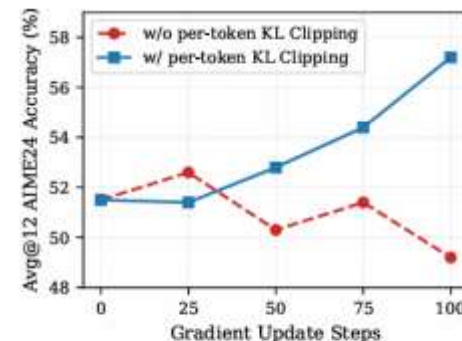


Figure 4. Effect of Per-Token pointwise KL Clipping on Qwen3-1.7B evaluated on AIME24. Clipping prevents performance collapse.

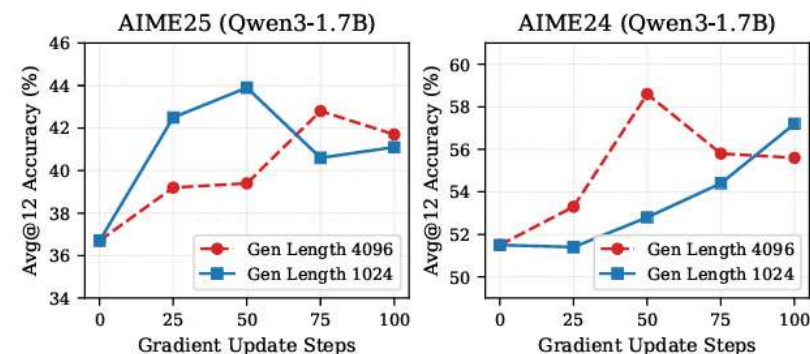


Figure 5. Effect of Generation Length on Qwen3-1.7B. We compare student generation length of 1024 vs 4096 on AIME25 and AIME24.

□ Vision-OPD:

[\[2605.18740\] Vision-OPD: Learning to See Fine Details for Multimodal LLMs via On-Policy Self-Distillation](#)

Vision-OPD: Learning to See Fine Details for Multimodal LLMs via On-Policy Self-Distillation

Qianhao Yuan^{1,2}, Jie Lou³, Xing Yu³, Hongyu Lin¹, Le Sun¹, Xianpei Han¹, Yaojie Lu¹

□ Motivation:

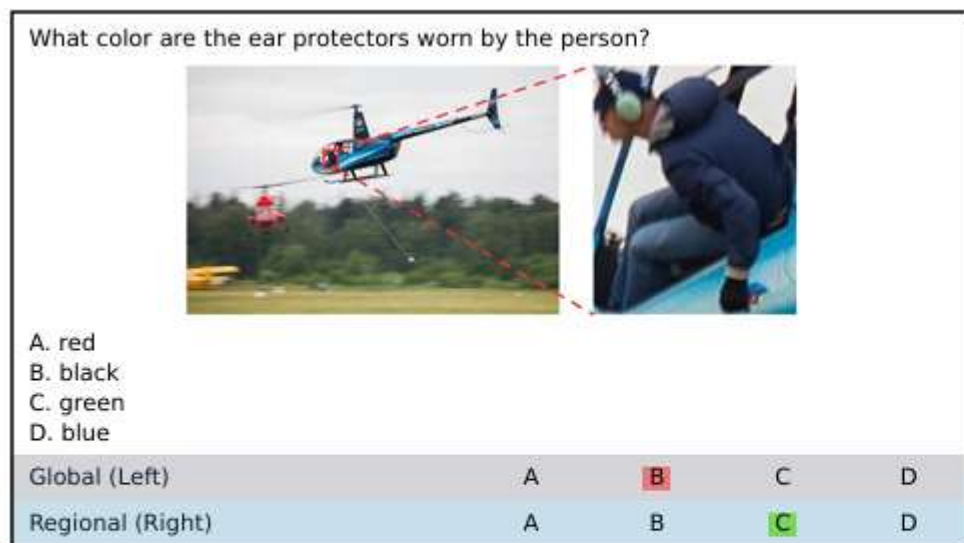


Figure 2: A case of the regional-to-global gap, based on Qwen3.5-9B. The global image input leads to the wrong answer, while the cropped region input yields the correct answer.

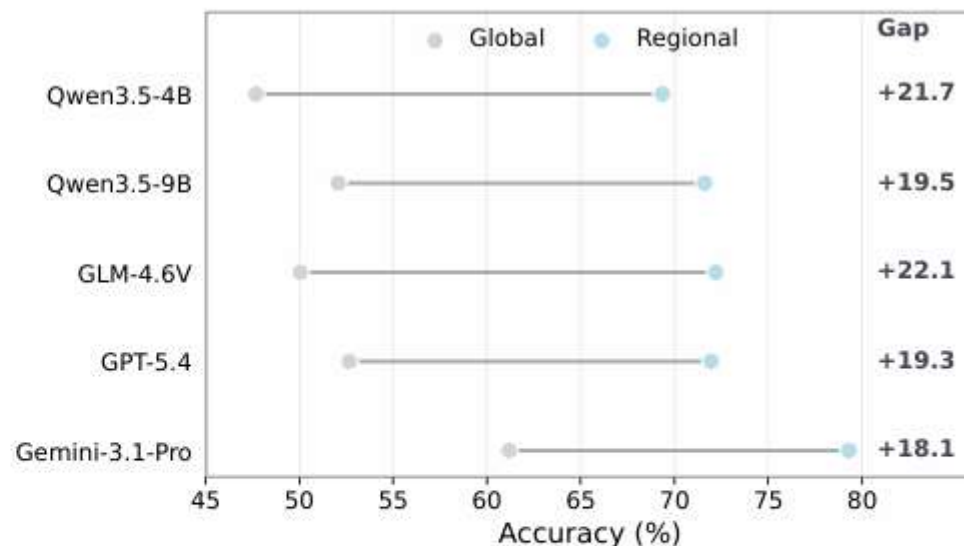
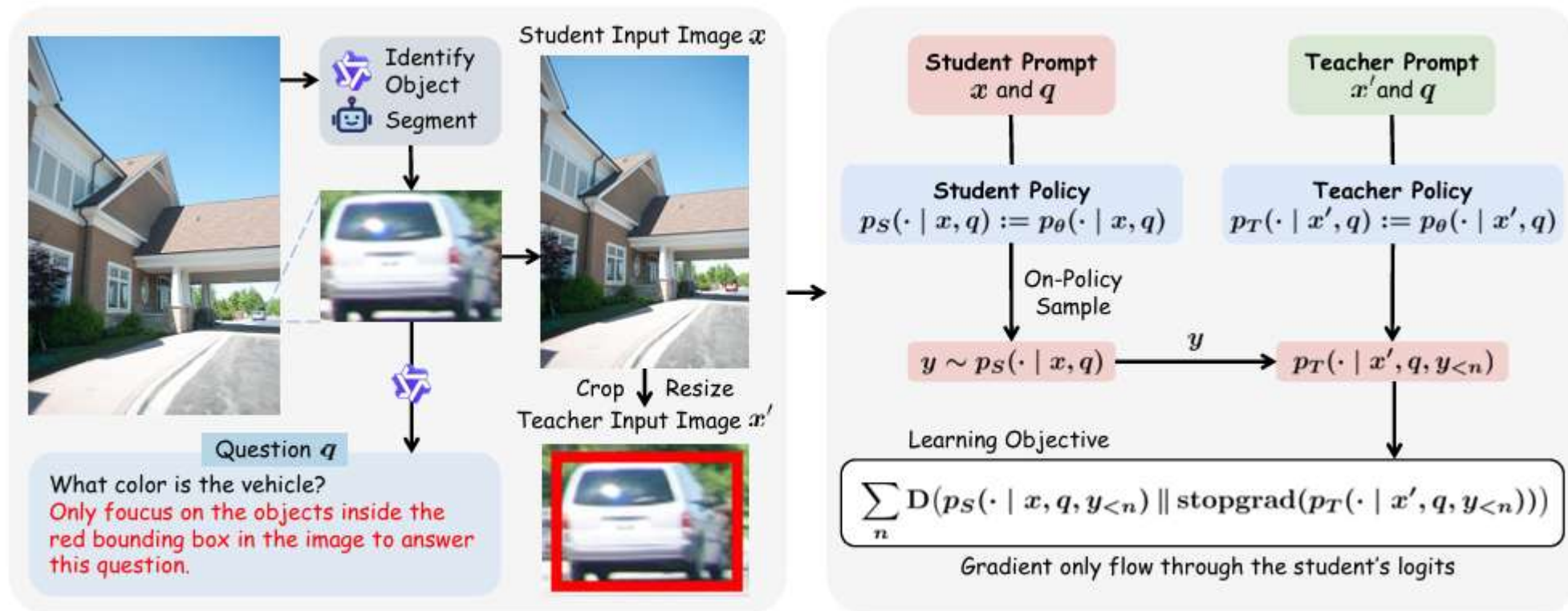


Figure 3: The accuracy of regional inputs is consistently higher than the accuracy of global inputs, revealing a persistent regional-to-global gap across multiple MLLMs.

□ Method

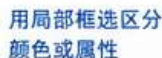


□ Main Result:

Table 1: Comparison with SOTA MLLMs. We report accuracy (%) for each model. Among open-source models (single forward pass), the best results are highlighted in **bold**, and the second-best are underlined. Vision-OPD achieving the best overall performance.

Model	Param Size	V* Bench	ZoomBench	HR-Bench 4K	HR-Bench 8K	MME-RW EN	MME-RW CN	Average
<i>“Thinking-with-Images” Agentic Models</i>								
DeepEyes	7B	85.86	46.51	75.13	72.63	64.10	64.09	68.05
Thyme	7B	82.20	45.09	77.00	72.00	64.80	64.59	67.61
DeepEyesV2	7B	81.68	44.97	77.88	73.75	64.90	65.07	68.04
SenseNova-MARS	8B	92.15	47.81	83.13	78.38	67.90	68.90	73.05
<i>Closed-Source Models (Single Forward Pass)</i>								
GPT-5.2	-	79.06	50.89	81.12	78.38	72.60	68.80	71.81
GPT-5.4	-	76.96	52.66	84.00	77.88	74.20	70.93	72.77
Gemini-3.1-Pro	-	87.96	61.18	89.63	86.88	76.53	73.31	79.25
Gemini-3.5-Flash	-	89.01	61.42	89.12	86.62	75.31	73.97	79.24
<i>Open-Source Models (Single Forward Pass)</i>								
Qwen3.5	4B	84.29	47.69	84.38	80.13	63.86	63.70	70.68
Qwen3.5	9B	82.72	52.07	85.75	80.63	71.40	67.67	73.37
MiMo-VL-RL	7B	83.25	45.68	73.50	69.38	62.73	55.89	65.07
Qwen3-VL-Instruct	8B	84.82	42.96	79.63	75.25	63.19	64.61	68.41
ZwZ	8B	87.96	56.69	83.63	81.75	66.57	68.09	74.12
MiniCPM-V-4.5	9B	70.68	42.60	69.63	61.50	62.65	61.64	61.45
GLM-4.6V	106B	86.91	50.06	82.13	78.88	65.57	65.62	71.53
Qwen3-VL-Instruct	235B	91.10	56.09	86.13	80.38	71.74	69.04	75.75
Qwen3.5	397B	87.96	57.16	89.38	85.50	<u>74.82</u>	69.82	<u>77.44</u>
Kimi-K2.6	1T	88.48	53.14	81.88	78.00	69.22	66.13	72.81
Vision-OPD (Ours)	4B	<u>92.15</u>	<u>59.76</u>	84.50	80.38	74.88	70.76	77.07
Vision-OPD (Ours)	9B	94.76	65.80	<u>88.13</u>	85.50	73.40	<u>70.46</u>	79.68

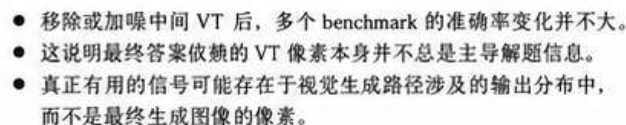
3. 关键观察：有用的 VT 只是生成出来的像素



这些图像不引入新标注，也不是用户输入，而是模型推理过程中自己生成的秘密中间状态。



- ## 5. 实验结果：更快且更准的纯文本推理



Parameter Optimization

- 使用 Teacher (VT 为主) 提供中间分布轨迹, 作为学生模型的 **privileged trace**。
- 通过 JSD 蒸馏, 在每个解码步对齐学生分布 P_S 与教师分布 P_T , 其中 $M = \beta P_T + (1 - \beta) P_S$ 。
- 训练时 Teacher 不更新参数, 只用于提供路径。
- 推理时 Student 直接输出答案, 无需生成中间图像。

Method	VSP $\uparrow$	VisPuzzle $\uparrow$	ChartQA $\uparrow$	VStar $\uparrow$	BLINK-J $\uparrow$
Vision-Language Models (VLMs)					
InternVL3.5-8B	8.17	34.75	76.26	68.59	71.33
InternVL3.5-8B	20.16	36.50	80.04	76.96	80.07
Qwen3-VL-8B	25.98	38.50	82.23	85.62	80.89
Qwen3-VL-32B	40.66	75.23	93.48	88.94	94.66
GPT-4o	40.00	36.35	78.93	93.38	88.00
GPT-5	33.03	77.50	78.06	75.33	77.38
Unified Multimodal Models (UMMs)					
Janus-Pro-7B	0.00	33.50	43.08	38.22	50.67
Chameleon-7B	0.00	38.50	1.03	30.23	8.09
HAEL-7B	0.00	28.00	23.03	32.83	89.08
ThinkMorph	75.83	77.50	78.00	67.01	66.00
Ours					
Text-only SFT	49.17	63.50	81.66	56.02	68.67
Visual-OPSD-Noise	50.83	64.50	73.77	61.70	68.66
Visual-OPSD (Ours)	55.83	86.00	78.79	64.92	77.33
Δ vs. ThinkMorph	+10.00	+8.50	+6.79	-2.09	+11.33
Δ vs. Text-only SFT	+36.66	+22.50	-2.87	+5.90	+8.66

Method	MMVP $\uparrow$	SAT $\uparrow$	BLINK $\uparrow$	CV-Bench $\uparrow$	Avg $\uparrow$	Lat. (s)
Vision-Language Models (VLMs)						
InternVL3.5-8B	76.33	45.33	59.60	81.99	58.04	-
InternVL3.5-8B	80.33	49.33	62.65	85.96	63.67	-
Qwen3-VL-8B	77.66	54.00	69.43	85.03	64.58	-
Qwen3-VL-32B	80.00	76.00	66.22	87.51	71.44	-
GPT-4o	84.67	28.09	60.28	75.61	59.62	-
GPT-5	86.23	73.50	69.86	85.48	75.58	-
Unified Multimodal Models (UMMs)						
Janus-Pro-7B	22.00	38.51	67.83	39.68	-	-
Chameleon-7B	47.67	10.67	16.52	36.52	19.71	-
HAEL-7B	70.33	44.67	47.66	76.03	51.02	-
ThinkMorph	78.33	52.67	59.49	80.86	70.63	142.8
Ours						
Text-only SFT	76.33	46.63	54.39	77.37	63.75	28.5
Visual-OPSD-Noise	75.33	48.00	55.49	79.09	64.15	14.9
Visual-OPSD (Ours)	77.33	54.00	61.44	80.64	74.03	10.0
Δ vs. ThinkMorph	-1.00	+1.53	+1.95	-0.22	+3.40	143.8
Δ vs. Text-only SFT	+1.00	+7.37	+7.05	+3.27	+10.28	2.99

- 评估包含 9 个 benchmark:
VSP、VisPuzzle、ChartQA、
V*、BLINK-J、MMVP、
SAT、BLINK、CV-Bench。
- Visual-OPSD 在 6/9 个
benchmark 上超越了 VT
生成的 teacher。
- 相对 ThinkMorph teacher:
平均提升 **+3.40pp**。
- 相对 text-only SFT:
平均提升 **+10.28pp**。
- 推理延迟:
10.0 秒/样本 vs.
142.8 秒/样本。



Visual Thought 的价值不一定在最终生成出来的图像像素，而在**视觉生成路径涉及的推理分布**。Visual-OPSD 将这种 VT 条件的 privileged distribution 蒸馏到纯文本 Student 中，从而保留空间推理能力，同时**大幅提升推理效率和真实的部署表现**。



西安交通大学
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谢谢大家!

精勤求學
敦篤勵志
果毅力行
忠恕任事